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Nitte (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, December 2025

25EEE103 – BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

(For AER, CIV, MEC)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing One full question from **each Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

CO1: Q1 to Q4; CO2: Q5 to Q8; CO3: Q9 to Q12; CO4: Q13 to Q16; CO5: Q17 to Q20.

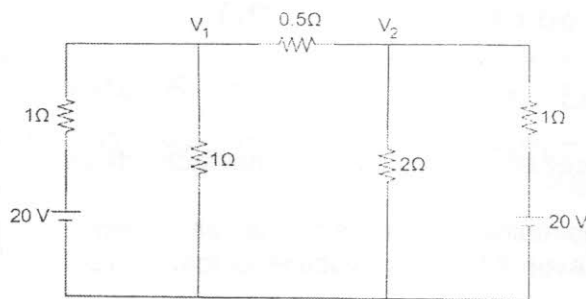
1. Reactive power is associated with which component?
A) Resistor
B) Transformer core
C) Inductor or Capacitor
D) Battery
2. In a purely resistive AC circuit, the power factor is:
A) 0
B) 1
C) 0.5
D) -1
3. If the time period of a waveform is 20 ms, its frequency is:
A) 20 Hz
B) 100 Hz
C) 50 Hz
D) 200 Hz
4. ----- energy (in kilowatthours) is required to light a 60W bulb continuously for 1 year (365 days)
A) 325.6kWh
B) 525.6kWh
C) 425.6kWh
D) 625.6kWh
5. Which motor is preferred for constant speed applications?
A) DC Series motor
B) DC Compound motor
C) DC Shunt motor
D) Universal motor
6. The rotor of a 3-phase induction motor always rotates at
A) Less than synchronous speed
B) More than synchronous speed
C) Synchronous speed
D) Zero speed
7. A transformer is an efficient device as it is -----
A) magnetically coupled
B) electrically coupled
C) all the above
D) static device
8. A 2000/200V, 20kVA ideal transformer has 60 turns in the secondary. The number of primary turns is -----
A) 6
B) 60
C) 600
D) 6000
9. In the piecewise linear model, the diode is represented by
A) Threshold voltage and forward resistance
B) Current source
C) Capacitor
D) Inductor
10. The ripple factor for a half-wave rectifier is
A) 0.48
B) 1.21
C) 0.33
D) 1
11. The ideal diode conducts with a voltage drop of
A) 0.3 V
B) 1
C) -0.3 V
D) 0

12. The input to a HWR is $v=200\sin 50t$. If $R_L=1\text{ k}\Omega$ and forward resistance of the diode is $50\text{ }\Omega$, then peak value of the current will be -----
- A) 0.15 B) 0.19
C) 2.2 D) 3.4
13. MOSFET is a----- device
- A) current controlled B) voltage controlled voltage source
C) voltage controlled D) voltage controlled current source
14. In the ohmic region, the MOSFET can be used as
- A) Voltage variable inductor B) Voltage variable resistor
C) Voltage variable capacitor D) Voltage variable current
15. The DC load line of BJT represents:
- A) The relationship between the AC input and output in a transistor circuit B) The relationship between the collector current (I_C) and collector-emitter voltage (V_{CE}) for a given circuit.
C) The maximum power dissipation of the transistor D) The frequency response of the transistor circuit.
16. A transistor has the base current of 0.06 mA and an emitter current of 9.00 mA . Then the value of alpha is -----
- A) 9.9 B) 1.01
C) 0.99 D) 900
17. The output voltage V_0 of an Op Amp differentiator is given by
- A) $V_0 = -\frac{1}{C_f} \int_0^t V_{in} dt$ B) $V_0 = -\frac{1}{R_f C} \frac{dV_{in}}{dt}$
C) $V_0 = -R_f C \frac{dV_{in}}{dt}$ D) $V_0 = -\frac{1}{R_f} \int_0^t V_{in} dt$
18. Which is not the ideal characteristic of an Op-Amp?
- A) Input resistance is zero B) Bandwidth is infinity
C) Output impedance is zero D) Open loop voltage gain is infinity
19. The duty cycle 'D' of IC 555 timer in astable mode in terms of resistors used in the external circuit is defined by
- A) $\frac{R_1 + R_2}{R_1 + 2R_2}$ B) $\frac{R_1 - R_2}{R_1 - 2R_2}$
C) $\frac{R_1 + 2R_2}{R_1 + R_2}$ D) $\frac{R_1 + R_2}{R_1 - 2R_2}$
20. A sine wave of 0.5 V peak voltage is applied as an input to a non-inverting amplifier with $R_i=12\text{ k}\Omega$ and $R_f=24\text{ k}\Omega$. The output voltage is -----
- A) -2.5 V peak B) 1.5 V peak
C) 10 V peak D) -5 V peak

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Real power, reactive power, apparent power and power factor.	4	L*1	1	1
	b) Define average value of an alternating quantity, also derive an expression for average value of current	6	L2	1	1,2

- c) Using Nodal analysis, calculate the voltages V_1 and V_2 in the circuit of figure 1.c



6 L3 1 2

2. a) Define
- Form factor
 - Peak factor
 - Power and
 - Energy
- b) Show that current through the pure capacitor lags behind the voltage across it by 90° . Draw the vector diagram.
- c) A circuit having resistance of $6\ \Omega$, an inductance of $0.075\ \text{Henry}$ and a capacitance of $50\ \mu\text{F}$ in series is connected across a 100V , 50Hz supply. Calculate :
- the impedance
 - the current
 - the voltage across R, L and C
 - the phase difference between the current and the supply voltage.

4 L1 1 1

6 L2 1 1,2

6 L3 1 2

Unit – II

3. a) Identify the differences between the squirrel cage and wound rotor three phase induction motors.
- b) Discuss the characteristics and applications of DC series and shunt motors
- c) A eight-pole lap-connected $230\ \text{V}$ shunt motor has 576 armature conductors. It takes $40\ \text{A}$ on full load. The flux per pole is $0.04\ \text{Weber}$. The armature and field resistances are $0.1\ \text{ohm}$ and $110\ \text{ohm}$ respectively. Contact drop per brush = $1\ \text{V}$. Determine the speed of the motor at full load.
4. a) Explain the construction of single phase transformers
- b) Derive the EMF equation of a single phase transformer.
- c) A single phase $50\ \text{Hz}$ transformer has 80 turns on the primary winding and 400 turns on the secondary winding. The net cross sectional area of the core is $200\ \text{cm}^2$. If the primary winding is connected to $230\ \text{V}$, 50Hz supply, determine:
- The e.m.f. induced in the secondary winding
 - Maximum value of the flux density in the core

4 L2 2 1

6 L2 2 1

6 L3 2 2

4 L2 2 1

6 L2 2 1,2

6 L3 2 2

Unit – III

5. a) What is meant by forward biasing and reverse biasing a diode?
- b) With neat circuit diagram, explain the operation of full wave rectifier and show that its efficiency is 81.2%

4 L1 3 1

6 L2 3 1,2

c)	A diode in a HWR whose internal resistance is $20\ \Omega$ is to supply power to a $1\ \text{k}\Omega$ load from a 20V , $50\ \text{Hz}$ supply, Determine: i. Peak load Current ii. DC load Current iii. RMS Load Current iv. Efficiency	6	L3	3	2
6.	a) Draw the diode equivalent circuit and also list out on what factor diode current depends.	4	L1	3	1
	b) Explain how a Zener diode maintains a constant output voltage when used as a voltage regulator, even if the input voltage or load current varies.	6	L2	3	1
	c) In a FWR, the load resistance is $1.5\text{k}\Omega$. Each diode has a forward resistance of 6Ω . The voltage across each half of the secondary winding is 220V . Find the ripple factor and rectification efficiency.	6	L3	3	2
Unit – IV					
7.	a) Compare enhancement and depletion type MOSFET	4	L1	4	1
	b) Explain the construction and operation of a BJT	7	L2	4	1
	c) Calculate I_E in a transistor for which $\beta=40$ and $I_B=30\mu\text{A}$	5	L3	4	2
8.	a) Draw the structure and symbol of Enhancement type MOSFET.	4	L1	4	1
	b) Illustrate the operation of CMOS inverter.	6	L2	4	1
	c) A transistor has $I_B = 0.08\text{mA}$ and $I_E = 9.60\text{mA}$. Determine its collector current I_C , α , and β .	6	L3	4	2
Unit – V					
9.	a) Draw the pin configuration and mention the functions of the same of a 555 timer IC.	4	L1	5	1
	b) Explain the working of a non-inverting amplifier and hence derive its gain.	6	L2	5	1,2
	c) For an IC timer based astable multivibrator, the duty cycle is $D=75\%$ with $f=1\text{kHz}$, $R_2=3.6\text{k}\Omega$ and $C=0.1\mu\text{F}$. Calculate the ON period T_{ON} and the value of R_1 .	6	L3	5	2
10.	a) Draw the equivalent circuit and list out any 4 ideal characteristics of an Op-Amp .	4	L1	5	1
	b) Describe the working of 78XX series IC voltage regulators.	6	L2	5	1,2
	c) In a inverting amplifier if $R_f=500\text{k}\Omega$, $R_i=100\text{k}\Omega$, Calculate output voltage for input of $0.4\sin\omega t$. Draw the input output waveform.	6	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
